

Titanic Dataset: Summary of Findings

1. Survival Rate by Gender

Females had a much higher survival rate compared to males.

Gender is a strong predictor of survival.

2. Passenger Class Impact

Passengers in 1st class had a significantly better chance of survival.

Higher class often indicates higher socio-economic status, which helped in rescue priority.

3. Age Influence

Younger passengers, especially children, had higher survival rates.

Age has a weak but visible correlation with survival.

4. Fare Correlation

Passengers who paid higher fares had a better chance of survival.

Indicates that wealthier passengers had better access to lifeboats or help.

5. Missing Data

Cabin, Age, and Embarked have missing values and require handling.

Cabin has the most missing values and might be dropped if necessary.

6. Feature Correlation

Moderate correlation found between:

- Fare and Pclass
- Pclass and Survival

Most other features are weakly correlated, which is typical in classification problems.